SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 50%

Duration: 1 Hour

QUESTIONS: 5 **Total Marks:50**

Annexure A

Scenario Based Assignment

Code of Conduct:

I Mohamed Aslam H (192572147) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Mohamed Aslam H

Annexure A

Scenario Based Assignment

1. Scenario : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty

ii. Explain how A* would handle the same situation, including how it balances cost and heuristic

iii. Analyze the impact of heuristic accuracy on both algorithms

iv. Discuss how dynamic changes (blocked paths) affect search performance

v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety

2. **Scenario**: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

i. Explain how Hill Climbing would work in this scenario and identify its limitations

ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation

iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations

iv. Recommend the best approach and justify your choice based on scalability and adaptability

3. **Scenario:** An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

i. Explain why this problem requires an online search agent instead of offline search

ii. Analyze the challenges of partial observability and dynamic environments

iii. Describe how the rover builds and updates its internal model

iv. Compare online search with uninformed and informed search strategies

v. Justify the most suitable approach considering uncertainty, adaptability, and safety

4. **Scenario:** A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

i. Clearly define variables, domains, and constraints with explanation

ii. Explain how backtracking search works in this scenario

iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency

iv. Analyze real-world challenges and justify the best strategy for scalability

5. **Scenario:** Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must complete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

i. Explain how the Minimax algorithm models this problem

ii. Analyze the computational challenges of large game trees

iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning

iv. Design an evaluation function and justify the factors considered

v. Recommend a strategy for balancing optimality and real-time performance

ANSWERS

1. AI-Powered Drone for Emergency Medicine Delivery

i. Greedy Best-First Search (GBFS)

Greedy Best-First Search selects the path that appears closest to the goal using only the heuristic value h(n) (straight-line distance).

Working:

- Chooses the node with the smallest heuristic value.
- Ignores the actual travel cost.
- Quickly reaches the destination if the heuristic is accurate.

Strengths

- Very fast.
- Uses less memory.
- Suitable for urgent situations where quick decisions are required.

Weaknesses

- Does not guarantee the shortest or safest path.
- Can choose dangerous or blocked routes.
- May get trapped in dead ends if heuristic information is misleading.

ii. A* Search

A* combines:

- g(n) = Actual travel cost from the start.
- h(n) = Estimated cost to reach the goal.

Formula:

$f(n) = g(n) + h(n)$

Working

- Evaluates both travel cost and estimated remaining distance.
- Avoids expensive or risky routes.
- Finds the optimal route when the heuristic is admissible.

Advantages

- Produces optimal solutions.
- Handles varying movement costs.
- Better suited for changing terrain and weather.

iii. Impact of Heuristic Accuracy

For Greedy Best-First Search

- Accurate heuristic → Very fast and efficient.
- Poor heuristic → Wrong path selection and longer delays.

For A*

- Accurate heuristic → Faster search with fewer nodes.
- Less accurate heuristic → Still finds the optimal path but explores more nodes.

Thus, A* is less affected by heuristic errors than Greedy Search.

iv. Effect of Dynamic Changes (Blocked Paths)

Blocked roads force the algorithm to:

- Discard invalid paths.
- Recalculate alternative routes.
- Increase computation time.

Greedy Search

- Frequently selects blocked paths because it only follows the heuristic.

A*

- Updates path costs and searches for better alternatives.
- More reliable during unexpected changes.

v. Recommendation

Recommended Algorithm: A*

Reasons

- Produces optimal paths.
- Considers both distance and travel cost.
- Handles weather and terrain variations.
- Provides safer navigation.
- Better performance under uncertainty.

Although Greedy Search is faster, A* offers the best balance between speed, safety, and optimality.

2. Smart Traffic Signal Optimization

Optimization Problem

Objective:

- Minimize waiting time.
- Minimize fuel consumption.
- Reduce traffic congestion.

Variables:

- Signal timings at every intersection.

Constraints:

- Pedestrian crossing times.
- Emergency vehicle priority.
- Traffic regulations.

Why Traditional Search is Inefficient

Traditional search explores every possible signal combination.

With hundreds of intersections, the search space becomes enormous (combinatorial explosion), making exhaustive search impractical.

i. Hill Climbing

Hill Climbing starts with an initial signal schedule and continuously changes timings to improve traffic flow.

Advantages

- Simple.
- Fast.
- Uses little memory.

Limitations

- Can become trapped in local optimum.
- Cannot guarantee the global best solution.
- Sensitive to starting condition

ii. Local Maxima, Plateaus and Ridges

Local Maximum

A solution better than nearby solutions but not globally optimal.

Example
Traffic improves in one area while congestion worsens elsewhere.

Plateau

Many neighboring solutions have the same value.

Example
Changing signal timing by one second has no noticeable effect.

Ridge

The optimal path requires several coordinated changes.

Example
Multiple intersections must change simultaneously to improve traffic.

Hill Climbing struggles because it changes only one variable at a time.

iii. Simulated Annealing and Genetic Algorithms

Simulated Annealing

- Occasionally accepts worse solutions.
- Escapes local maxima.
- Probability decreases as temperature cools.

Advantages:

- Avoids getting stuck.
- Explores more solutions.

Genetic Algorithm

Inspired by natural evolution.

Steps:

1. Generate population.
2. Evaluate fitness.
3. Select best solutions.
4. Perform crossover.
5. Apply mutation.
6. Repeat.

Advantages:

- Searches many solutions simultaneously.
- Handles large search spaces.
- Highly adaptable.

iv. Recommendation

Best Algorithm: Genetic Algorithm

Reasons:

- Highly scalable.
- Suitable for dynamic traffic.
- Finds near-optimal solutions.
- Easily adapts to changing traffic conditions.
- Better than Hill Climbing for large optimization problems.

3. Mars Autonomous Rover

i. Why Online Search?

Offline search requires a complete map before moving.

Mars exploration has:

- Unknown terrain.
- Hidden obstacles.
- Limited sensor range.

Therefore, the rover must:

- Observe.
- Decide.
- Move.
- Learn continuously.

This makes Online Search necessary.

ii. Challenges

Partial Observability

The rover cannot see the entire environment.

Example:
Hidden craters beyond sensor range.

Dynamic Environment

Conditions change due to:

- Dust storms.
- Loose rocks.
- Slopes.

The rover must frequently update its decisions.

iii. Internal Model

The rover maintains an internal map.

It stores:

- Safe paths.
- Obstacles.
- Visited locations.
- Sample collection points.

As new sensor data arrives, the map is updated.

This improves future navigation.

iv. Online Search vs Other Searches

Online Search	Uninformed Search	Informed Search
Learns while moving	Needs complete map	Uses heuristic
Works in unknown environments	Suitable only for known environments	Requires reliable heuristic
Adaptive	Less flexible	Faster than uninformed

v. Recommendation

Best Approach: Online Search Agent

Reasons:

- Handles uncertainty.
- Learns continuously.
- Updates knowledge in real time.
- Safe navigation.
- Ideal for Mars exploration.

4. University Exam Timetable (CSP)

CSP Formulation

Variables

Each exam represents one variable.

Example:

Math = X_1

Physics = X_2

Chemistry = X_3

Domains

Possible time slots and exam halls.

Example:

Monday Morning

Monday Afternoon

Tuesday Morning

etc.

Constraints

- No student has overlapping exams.
- Limited exam halls.
- Subject ordering.
- Faculty availability.
- Special accommodations.

i. Variables, Domain and Constraints

Variables:

- Exams.

Domains:

- Available time slots and rooms.

Constraints:

- No overlaps.
- Hall capacity.
- Faculty availability.
- Subject precedence.
- Accessibility requirements.

ii. Backtracking Search

Backtracking assigns one exam at a time.

If a conflict occurs:

- Undo previous assignment.
- Try another slot.

Continue until all exams are scheduled.

Advantages:

- Guarantees a valid solution if one exists.

ii. Forward Checking

Forward Checking removes impossible values from future variables.

Example:

If Math is fixed on Monday Morning,

Physics cannot use the same slot for common students.

Benefits:

- Reduces unnecessary searching.
- Detects conflicts early.
- Improves efficiency.

iv. Real-world Challenges

Challenges:

- Thousands of students.
- Hundreds of exams.
- Multiple constraints.

- Last-minute changes.

Best Strategy

Use:

- Backtracking
- Forward Checking
- Constraint Propagation
- MRV (Minimum Remaining Values)
- Degree Heuristic

This combination improves scalability and reduces computation time.

5. Strategic Game AI (Minimax & Alpha-Beta Pruning)

i. Minimax Algorithm

Minimax assumes:

- AI tries to maximize its score.
- Human tries to minimize the AI's score.

The algorithm explores possible future moves.

It chooses the move with the best guaranteed outcome.

ii. Computational Challenges

Game trees grow exponentially.

Example:

If branching factor = 30

Depth = 8

Nodes explored:

$30^8 \approx 656$ billion nodes.

Problems:

- Huge memory usage.
- Long computation time.

- Not suitable for real-time decisions without optimization.

iii. Alpha-Beta Pruning

Alpha-Beta improves Minimax by removing branches that cannot influence the final decision.

Alpha (α)

Best value for the maximizing player.

Beta (β)

Best value for the minimizing player.

If

$\alpha \geq \beta$

the remaining branch is pruned.

Benefits:

- Explores far fewer nodes.
- Produces the same optimal move as Minimax.
- Faster response.
- Suitable for real-time games.

iv. Evaluation Function

Example evaluation score:

Evaluation = 5 × Resources + 10 × Army Strength + 15 × Territory Control + 20 × Objective Progress − 10 × Damage Taken

Factors:

- Army strength.
- Available resources.
- Territory controlled.
- Health.
- Remaining objectives.
- Enemy strength.
- Strategic position.

A higher score indicates a better game state.

v. Recommendation

Best Strategy: Minimax + Alpha-Beta Pruning

Reasons:

- Maintains optimal decisions.
- Significantly reduces computation.
- Works well under strict time limits.
- Suitable for complex real-time strategy games

Conclusion

Question	Best Algorithm
Emergency Drone Routing	**A*** Search
Smart Traffic Optimization	**Genetic Algorithm**
Mars Rover Navigation	**Online Search Agent**
Exam Timetable Scheduling	**Backtracking + Forward Checking (CSP)**
Strategy Game AI	**Minimax + Alpha-Beta Pruning**

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand